

JIRAWAT WONGMAHASARN

MES Software Engineer | Siemens Limited

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Digital Manufacturing Specialist with strong experience in OT/IT integration, MES implementation, and Industry 4.0 adoption. Adept at driving digital transformation initiatives through smart automation, IoT connectivity, and data analytics across manufacturing and logistics environments. Experienced in leading cross-functional collaboration between engineering, production, and IT teams to deliver impactful, scalable digital solutions.

Skills & Proficiencies

- **Technical Skills:** C#, C/C++, Python, SQL, HTML5, JavaScript, CSS, Bootstrap, Microsoft SQL Server, SSRS, MQTT, HTTP, RTOS, IoT, Edge Computing
- **Tools & Technologies:** Visual Studio, WPF, Node-RED, Platform IO
- **Soft Skills:** Team Collaboration, Project Management, Resilience, Documentation

Experiences

- **Siemens Limited (July 2025 - Present): MES Software Engineer**

- Develop and implement Manufacturing Execution System (MES) and SCADA solutions using WinCC OA, writing custom scripts in C++ to automate workflows, build dynamic user interfaces, and streamline operational processes.
- Integrated MES software with PLC and SCADA systems using standard industrial communication protocols such as Modbus, OPC UA, and PROFINET, ensuring real-time data exchange between IT and OT layers.
- Collaborate with production, IT, and engineering teams to ensure seamless OT/IT data exchange and system interoperability aligned with ISA-95 architecture.
- Design and query relational databases using SQL to manage secure data processing, extract operational metrics, and engineer specialized reporting programs

- **Pteris Global Limited (April 2023 - June 2025): HLC Engineer**

- Designed and implemented .NET- and SQL-based control and monitoring applications for international airport Baggage Handling Systems (BHS), improving real-time system visibility and operational uptime.
- Integrated automation equipment with centralized control software, bridging OT-level devices with IT databases and reporting systems.
- Designed and implemented MSSQL Server databases and stored procedures, improving data flow between subsystems and enhancing traceability.
- Delivered pre-sale consultations and post-sale technical support, collaborating with global teams and clients to ensure smooth project execution.
- Authored detailed technical documentation and user manuals, and provided end-user training and workshops to ensure successful customer adoption.
- On-site collaboration with engineering and IT teams at Cairo International Airport to ensuring smooth deployment and integration with existing infrastructure.

Experiences

- **Advanced Info Service** (November 2021 – April 2023): **Embedded Software Engineer**
 - Developed MQTT and HTTP communication libraries in C/C++, enabling large-scale IoT device connectivity with the AIS IoT Platform (Magellan).
 - Designed device SDKs and developer tools to accelerate integration for AIS partners, contributing to scalable IoT ecosystems supporting smart manufacturing and agriculture applications.
 - Built C# and Python-based MQTT QA tools to test and validate OTA (Over-the-Air) communication reliability and performance.
 - Supported pre-sale solution design and provided post-sale integration assistance for enterprise clients, aligning with digital transformation and connectivity goals.
 - Authored comprehensive user guides and delivered technical workshops for partner enablement.
- **Belton Industrial (Thailand) Co.,LTD.** (January 2021 – November 2021): **System Analyst (IoT Developer)**
 - Designed and implemented IoT-based monitoring systems connecting machine-level data to real-time dashboards for factory performance tracking.
 - Developed multi-tasking C/C++ applications for IoT nodes (ESP32/Arduino) and web dashboards using HTML5, JavaScript, and SQL for production visibility.
 - Defined and implemented the data architecture for factory monitoring, improving machine utilization and process traceability.
 - Collaborated with production engineers and IT teams to align IoT solutions with digital transformation goals, enabling smart manufacturing initiatives.
- **Johnson and Johnson (Thailand) Ltd.** (September 2020 – December 2020) **Maintenance Process Engineer**
 - Improved manufacturing line efficiency through automation and camera-based inspection system enhancements, contributing to defect reduction and data-driven maintenance improvements.
- **Cooperative Project with SCG** (Jun 2019 - Aug 2019): **Engineer Intern**
 - Developed LoRa-based fleet monitoring systems for mining operations, collecting real-time data (speed, location, cycle time) for performance analysis.
 - Designed C# visualization software (WPF) displaying live operational metrics and route tracking an early application of industrial IoT and data-driven decision making.
- **Automation project** (2016 - 2017): Built a pneumatic automation control system using C# with MQTT integration, enabling remote PLC control through multi-device access (PC, tablet, mobile).

Education

King Mongkut's Institute of Technology Ladkrabang

Bachelor of Engineering in Manufacturing System Engineering

Bangkok, Thailand

June 2016 – June 2020